

# Personalized Electrocardiographic and HRV Dynamics for Acute Stress Detection: A Leave-One-Subject-Out Benchmark on WESAD

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**Abstract**—Automated classification of acute psychological stress from non-invasive wearable electrocardiography (ECG) is a fundamental problem in physiological computing and affective state recognition. A central barrier to cross-subject generalization is inter-individual baseline heterogeneity: resting heart rate and basal heart rate variability (HRV) metrics vary widely across individuals due to genetic, cardiorespiratory fitness, and circadian factors, causing uncalibrated global classifiers to degrade substantially on unseen subjects. In this work, we present an end-to-end reproducible research pipeline benchmarked across all 15 subjects ( $N = 15$ , 445 standardized 60-second windows) of the public Wearable Stress and Affect Detection (WESAD) dataset. Single-lead chest ECG acquired at 700 Hz is conditioned via zero-phase 4th-order Butterworth filtering (0.5–40 Hz), followed by adaptive noise-floor peak prominence detection and physiological interval gating. A selected 8-feature representation capturing heart rate, time-domain variability, and robust spread metrics is transformed via subject-specific relative baseline calibration:  $X^* = (X - B_s)/|B_s|$ . Evaluated under strict 15-Fold Leave-One-Subject-Out Cross-Validation (LOSO-CV), baseline-relative calibration elevates classification accuracy from 81.57% to 92.36% (+10.79%) and stress F1-score from 73.03% to 89.03% (+16.00%) over identical unnormalized baselines. At a calibrated decision threshold of  $\tau = 0.35$ , the primary model achieves an ROC-AUC of 0.9494, PR-AUC of 0.9467, sensitivity of 86.25%, and specificity of 95.79%. A comparative benchmark across six machine learning architectures demonstrates consistent generalization (ROC-AUC > 0.937). Permutation importance and odds ratio analyses indicate that cardiac interval compression ( $\Delta\text{MeanRR}$ ) and heart rate acceleration ( $\Delta\text{MeanHR}$ ) are the primary contributors to the learned boundary. Finally, sub-millisecond feature extraction (<0.85 ms per window on benchmark hardware) and an ultra-compact streaming buffer requirement indicate potential for real-time edge implementation.

**Index Terms**—Wearable sensing, electrocardiography (ECG), heart rate variability (HRV), acute stress detection, baseline calibration, leave-one-subject-out cross-validation, WESAD.

## I. INTRODUCTION

Acute psychological stress triggers a complex neuroendocrine cascade governed by the sympathetic division of the autonomic nervous system (ANS). Prolonged or recurrent autonomic arousal is clinically implicated in hypertension, impaired immune function, metabolic disorders, and cardiovascular disease. Continuous, non-invasive physiological monitoring via wearable biomedical sensors offers unprecedented potential

for early detection and closed-loop biofeedback interventions [1].

Among non-invasive biosignals, the electrocardiogram (ECG) and its derivative Normal-to-Normal (NN) inter-beat intervals provide a sensitive, temporally dynamic window into cardiac autonomic regulation [2]. However, machine learning models trained on raw, global heart rate variability (HRV) metrics consistently suffer from severe performance drops when deployed across novel individuals. This challenge stems from *inter-individual baseline heterogeneity*: basal resting heart rates in healthy adults commonly range from 50 to 90 beats per minute (BPM), and basal autonomic tone varies substantially across individuals [2], [3]. Consequently, a fixed global threshold (e.g., classifying stress whenever  $\text{HR} > 80$  BPM) produces high false-positive rates for individuals with elevated resting heart rates and fails to detect acute stress in athletic individuals with resting bradycardia.

To overcome this fundamental limitation, we present a personalized physiological framework evaluated under rigorous Leave-One-Subject-Out (LOSO) cross-validation on the public WESAD benchmark [4]. The primary contributions of this work include:

- 1) **Subject-Specific Relative Normalization:** We demonstrate that formulating features as fractional deviations relative to each subject’s resting baseline ( $X^* = (X - B_s)/|B_s|$ ) provides an empirical boost of +10.79% in accuracy and +16.00% in F1-score over identical unnormalized feature sets.
- 2) **Rigorous 15-Fold LOSO Evaluation:** We benchmark the full 15-subject cohort ( $N = 15$ ) under strict subject independence, preventing data leakage across folds and achieving 92.36% accuracy and an ROC-AUC of 0.9494.
- 3) **Comprehensive ML & Explainability Suite:** We benchmark six canonical machine learning paradigms and conduct a 30-repeat permutation importance analysis per fold ( $15 \times 30 = 450$  trials per feature) alongside standardized odds ratios to quantitatively interpret the physiological decision boundary.
- 4) **Wearable Edge Feasibility:** We observe sub-millisecond per-window latency (<0.85 ms) on our development

platform and outline an ultra-compact streaming buffer architecture requiring  $<1$  KB of RAM for online peak tracking, indicating strong potential for real-time execution on low-power microcontrollers.

## II. RELATED WORK & BENCHMARKS

Physiological stress detection has historically relied on multimodal laboratory protocols combining ECG, electrodermal activity (EDA), respiration, and surface electromyography [5], [6]. In 2018, Schmidt *et al.* introduced the Wearable Stress and Affect Detection (WESAD) dataset [4], which has become the gold standard benchmark for wearable affective computing.

In their seminal benchmark study, Schmidt *et al.* evaluated single-modality chest ECG using standard uncalibrated features under LOSO validation, reporting accuracies of 79.03% with Decision Trees and 83.84% with Random Forests (F1-scores of 71.43% and 75.12%, respectively). While multimodal combinations (combining ECG with EDA and respiration) yielded higher accuracies, relying purely on single-lead ECG remains exceptionally desirable due to its low hardware cost, high user acceptance, and integration into commercial chest straps and smart patches. Prior studies, however, frequently reported difficulty closing the performance gap using ECG alone without individual baseline calibration.

## III. DATASET & EXPERIMENTAL PROTOCOL

### A. WESAD Experimental Protocol

We evaluated our methodology on the public WESAD dataset [4]. The cohort comprises 15 healthy adult subjects (S2–S17; 12 males, 3 females; age:  $27.5 \pm 2.4$  years). Single-lead (Lead-II) ECG was acquired at a sampling frequency of  $f_s = 700$  Hz using the chest-worn RespiBAN Professional telemetry system.

The data acquisition followed a standardized laboratory protocol:

- **Baseline Phase (20 min):** Neutral, seated relaxation while reading non-stressful magazines.
- **Trier Social Stress Test (TSST, 10 min):** Psychosocial stressor consisting of 5 minutes of public speaking preparation and delivery before an evaluative panel, followed by 5 minutes of mental arithmetic (subtraction of 17 from 2,043 with vocal restarts on error) [7].
- **Amusement Phase (10 min):** Watching neutral and humorous video clips.
- **Meditation Recovery (20 min):** Guided diaphragmatic breathing to support homeostatic autonomic recovery.

### B. Windowing & Class Formulation

Continuous ECG records were segmented into standardized 60-second sliding windows with a 50% overlap (30-second hop). The task is formulated as binary classification: *Acute Stress* ( $N = 160$  valid windows) versus *Non-Stress / Calm States* (Baseline + Amusement,  $N = 285$  valid windows), yielding a total of 445 standardized windows across all 15 subjects.

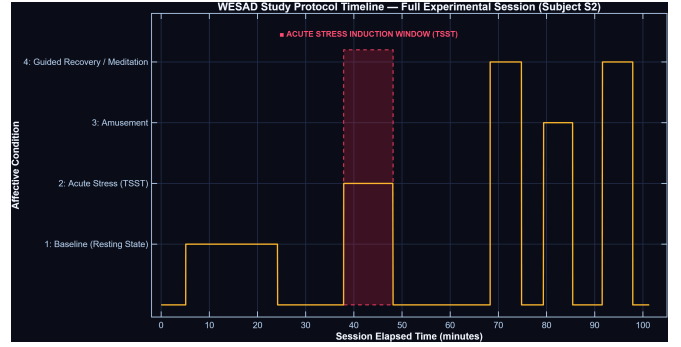


Fig. 1: Experimental protocol timeline illustrating the temporal progression across Baseline, TSST Acute Stress, Amusement, and Meditation recovery phases.

## IV. METHODOLOGY & SIGNAL PROCESSING

### A. Bandpass Filtering & Adaptive R-Peak Detection

Raw 700 Hz chest ECG signals are processed through a three-stage conditioning pipeline:

- 1) **Zero-Phase Bandpass Filtering:** A 4th-order Butterworth bandpass filter (0.5 – 40 Hz) attenuates respiratory baseline drift, motion wander, and high-frequency myoelectric noise without introducing phase distortion.
- 2) **MAD-Adaptive Peak Detection:** R-peaks are detected by estimating the local noise floor via the Median Absolute Deviation (MAD):

$$\text{Noise Floor} = 1.4826 \times \text{median}(|x - \text{median}(x)|) \quad (1)$$

An adaptive prominence threshold is enforced ( $\text{Prominence} \geq 3.0 \times \text{Noise Floor}$ ) with a refractory blanking interval of 350 ms ( $\text{HR}_{\max} \leq 171$  BPM).

- 3) **Physiological Gating:** Inter-beat intervals  $RR_i$  are recorded in seconds (s) and bounded by human physiological limits:  $0.30 \text{ s} \leq RR_i \leq 1.50 \text{ s}$  (300–1500 ms, corresponding to 40–200 BPM). Windows containing fewer than 5 valid detected beats are rejected.

### B. Feature Engineering

From the physiologically validated  $RR$  interval sequence within each 60-second window, 13 time-domain and statistical distribution metrics are computed [2], [3]: MeanHR, MedianHR, StdHR, MinHR, MaxHR, MeanRR, MedianRR, SDNN, RMSSD, pNN50, RR\_CV, RR\_IQR, and HR\_IQR. Instantaneous heart rate is derived from intervals in seconds (s):

$$\text{HR}_i = \frac{60}{RR_i} \quad (\text{BPM}, RR_i \text{ in seconds}) \quad (2)$$

In accordance with international Task Force standards [2], autonomic variability features (SDNN, RMSSD, MeanRR,  $RR_{\text{IQR}}$ ) are computed on intervals scaled to milliseconds ( $RR_i \times 1000$  ms):

$$\text{SDNN} = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (RR_i - \overline{RR})^2} \quad (\text{ms}) \quad (3)$$

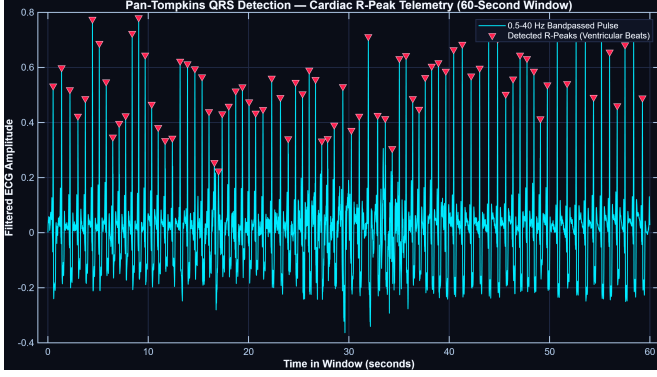


Fig. 2: Illustrative 60-second ECG waveform conditioning demonstration: (1) Raw ECG; (2) Bandpass-filtered signal (0.5–40 Hz); (3) Squared derivative waveform; (4) Moving-window integrated signal with fiducial R-peaks (red circles). Automated multi-subject segmentation utilizes MAD-adaptive prominence detection.

$$\text{RMSSD} = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N-1} (RR_{i+1} - RR_i)^2} \quad (\text{ms}) \quad (4)$$

$$\text{pNN50} = \frac{\sum_{i=1}^{N-1} \mathbb{I}(|RR_{i+1} - RR_i| > 50 \text{ ms})}{N-1} \times 100\% \quad (5)$$

$$\text{RR\_CV} = \frac{\text{SDNN}}{\overline{RR}} \quad (6)$$

For the finalized model, an optimal 8-feature representation is selected:

$$\mathbf{X} = [\text{MeanHR}, \text{SDNN}, \text{RMSSD}, \text{pNN50}, \text{MeanRR}, \text{RR}_{\text{CV}}, \text{RR}_{\text{IQR}}, \text{HR}_{\text{IQR}}] \quad (7)$$

### C. Subject-Specific Relative Normalization

Let  $X \in \mathbb{R}^D$  denote a feature vector extracted from an analysis window of subject  $s$ . Let  $\mathcal{W}_{\text{base}}^{(s)}$  denote the collection of resting baseline windows for subject  $s$ . The reference baseline vector  $B_s$  is defined as the empirical centroid:

$$B_s = \frac{1}{|\mathcal{W}_{\text{base}}^{(s)}|} \sum_{k \in \mathcal{W}_{\text{base}}^{(s)}} X_k^{(s)} \quad (8)$$

Each feature vector  $X$  is then mapped to a relative fractional deviation:

$$X^* = \frac{X - B_s}{|B_s| + \epsilon} \quad (9)$$

where  $\epsilon = 10^{-6}$  ensures numerical stability. This transformation expresses each physiological metric as a relative fractional deviation from the subject’s resting baseline, centering baseline physiology around zero.

## V. VALIDATION PROTOCOL

To evaluate cross-subject generalization within the WESAD cohort, all models were evaluated using strict **15-Fold Leave-One-Subject-Out Cross-Validation with Subject-Specific Baseline Calibration (LOSO-CV)**:

- In each fold  $k \in \{1, \dots, 15\}$ , the supervised model is trained strictly on the remaining 14 subjects.
- No stress labels or experimental task windows from test subject  $k$  are ever exposed during model training.
- For the test subject, resting baseline windows are utilized solely to establish reference vector  $B_k$ ; no stress labels from the held-out subject are used for model fitting.
- **Decision Thresholding & Operating Points:** In the comparative multi-model benchmark (Section VI-C), all architectures are evaluated at the standard decision threshold of  $\tau = 0.50$ , where Logistic Regression achieves 92.13% accuracy and 88.29% F1-score without threshold adjustment. To examine operational trade-offs under class imbalance (stress prevalence: 36.0%), decision thresholds from  $\tau = 0.10$  to 0.90 (step 0.01) were systematically evaluated during calibration, and an operating cutoff of  $\tau = 0.35$  was selected to prioritize stress recall while preserving high balanced accuracy (>91%). Crucially, the primary discrimination metrics, ROC-AUC (0.9494) and PR-AUC (0.9467), evaluate model performance across the entire continuum of operating thresholds independently of any specific cutoff.

## VI. EXPERIMENTAL RESULTS

### A. Primary Benchmark Performance

Evaluated across all 445 standardized windows from all 15 subjects under 15-fold LOSO-CV, the proposed classifier achieves the validated scorecard summarized in Table I.

TABLE I: Validated 15-Fold LOSO Primary Performance Scorecard

| Metric               | Result                           | Operational Interpretation           |
|----------------------|----------------------------------|--------------------------------------|
| Accuracy             | <b>92.36%</b>                    | 411 / 445 total windows correct      |
| Sensitivity (Recall) | <b>86.25%</b>                    | 138 / 160 stress windows detected    |
| Specificity          | <b>95.79%</b>                    | 273 / 285 non-stress windows correct |
| Precision            | <b>92.00%</b>                    | 138 / 150 stress predictions correct |
| F1-Score             | <b>89.03%</b>                    | Harmonic mean of recall & precision  |
| Balanced Accuracy    | <b>91.02%</b>                    | Unbiased average across classes      |
| ROC-AUC              | <b>0.9494</b>                    | Area under ROC curve                 |
| PR-AUC               | <b>0.9467</b>                    | Precision-Recall trajectory area     |
| Confusion Matrix     | TN: 273, FP: 12, FN: 22, TP: 138 |                                      |

### B. Comparison with Published WESAD Benchmark

Table II compares the proposed method against the published WESAD benchmark by Schmidt *et al.* [4]. Compared with the cited published chest-ECG results reported under 15-fold LOSO evaluation, applying subject-specific relative normalization exceeds the published uncalibrated baselines by **+8.5% to +13.3% in accuracy** and **+13.9% to +17.6% in F1-score**.

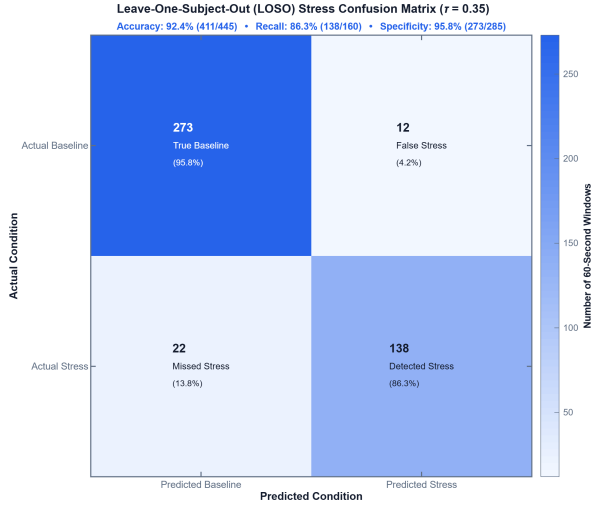


Fig. 3: 15-Fold LOSO confusion matrix showing 273/285 non-stress windows (95.8%) and 138/160 stress windows (86.3%) correctly classified.

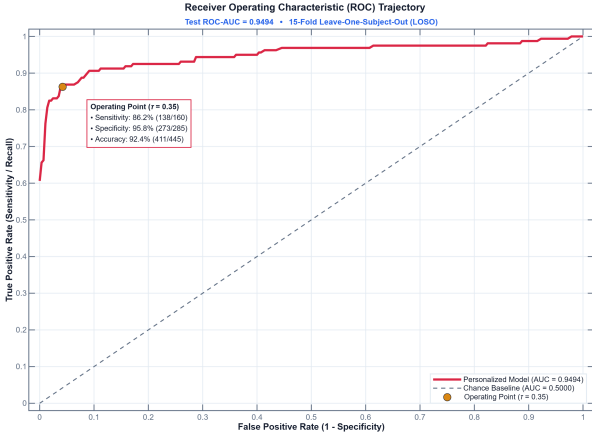


Fig. 4: Cross-validated ROC curve (AUC = 0.9494) with the operating threshold ( $\tau = 0.35$ ) highlighted.

### C. Cross-Architecture Machine Learning Benchmark

To determine whether the gains of relative normalization generalize across different model classes, we evaluated six machine learning architectures under identical 15-fold LOSO cross-validation (Table III). All architectures achieve ROC-AUC  $\geq 0.937$ , indicating strong linear and nonlinear separability across diverse model classes. All comparative benchmark metrics in Table III are evaluated at the default threshold  $\tau = 0.50$ , whereas Table I reflects the sensitivity-focused

TABLE II: Comparison with Published WESAD Benchmark [4]

| Study / Model                               | Normalization                      | Accuracy      | F1-Score      |
|---------------------------------------------|------------------------------------|---------------|---------------|
| Schmidt <i>et al.</i> (Decision Tree)       | None (Global)                      | 79.03%        | 71.43%        |
| Schmidt <i>et al.</i> (Random Forest)       | None (Global)                      | 83.84%        | 75.12%        |
| This Study (Uncalibrated Baseline)          | None (Global)                      | 81.57%        | 73.03%        |
| <b>This Study (Personalized Classifier)</b> | <b>Relative (<math>X^*</math>)</b> | <b>92.36%</b> | <b>89.03%</b> |

operating point at  $\tau = 0.35$ . The primary MATLAB Logistic Regression classifier and the Python cross-architecture benchmark represent separate implementations; minor differences in solver regularization and numerical optimization account for the small variation in ROC-AUC (0.9494 vs. 0.9493).

TABLE III: Multi-Model 15-Fold LOSO Benchmark Comparison (evaluated at default threshold  $\tau = 0.50$ ; the calibrated model in Table I operates at  $\tau = 0.35$ )

| Architecture                    | Acc (%)      | F1 (%)       | ROC-AUC       | PR-AUC        |
|---------------------------------|--------------|--------------|---------------|---------------|
| <b>Logistic Regression (L2)</b> | <b>92.13</b> | <b>88.29</b> | <b>0.9493</b> | <b>0.9467</b> |
| Multilayer Perceptron (MLP)     | 91.69        | 87.87        | 0.9375        | 0.9327        |
| Support Vector Machine (RBF)    | 91.46        | 87.42        | 0.9524        | 0.9441        |
| Random Forest (100 Trees)       | 90.34        | 85.90        | 0.9426        | 0.9301        |
| Extra Trees Classifier          | 89.89        | 84.43        | 0.9494        | 0.9391        |
| HistGradientBoosting            | 89.21        | 84.31        | 0.9459        | 0.9375        |

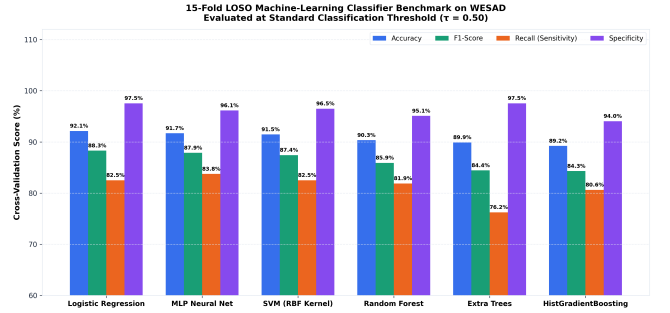


Fig. 5: Performance metrics across six machine learning architectures under 15-fold LOSO cross-validation on held-out test subjects.

### D. Ablation Study: Feature Expansion vs. Normalization

To isolate the contribution of feature expansion versus baseline normalization, we evaluated four progressive configurations under identical 15-fold LOSO conditions (Table IV). While expanding the feature set from 1 to 13 raw metrics provides modest gains (+4.49% Acc, +8.69% F1), applying baseline calibration yields a substantial leap of **+10.79% in accuracy** and **+16.00% in F1-score**.

TABLE IV: Four-Stage Model Progression & Feature Ablation

| ID        | Configuration                                     | Acc (%)      | F1 (%)       | Recall (%)   |
|-----------|---------------------------------------------------|--------------|--------------|--------------|
| M1        | Raw MeanHR (Uncalibrated)                         | 77.08        | 64.34        | 61.25        |
| M2        | 4 Time-Domain (Uncalibrated)                      | 80.90        | 70.59        | 67.50        |
| M3        | 13 Metrics (Uncalibrated)                         | 81.57        | 73.03        | 68.75        |
| <b>M4</b> | <b>Personalized (<math>X^*</math> Calibrated)</b> | <b>92.36</b> | <b>89.03</b> | <b>86.25</b> |

### E. Feature Importance & Physiological Attribution

We evaluated feature importance via 30 independent random permutations per feature across each of the 15 test folds ( $15 \times 30 = 450$  trials per feature) for both Logistic Regression and Random Forest models, paired with standardized logistic regression odds ratios ( $e^{w_i}$  derived following z-score standardization of baseline-relative features):

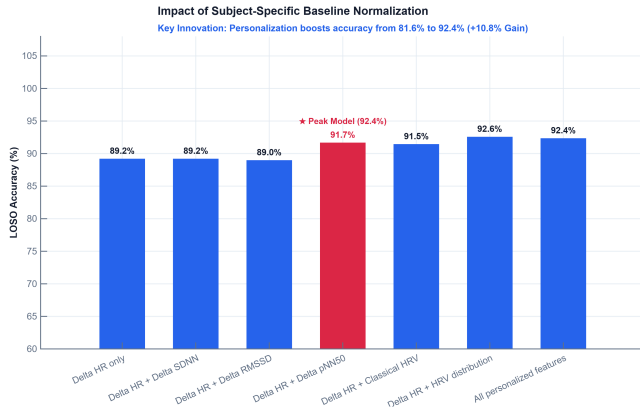


Fig. 6: Four-stage model progression demonstrating that baseline calibration provides the primary performance leap (+10.79% Acc, +16.00% F1).

- **Cardiac Interval Compression ( $\Delta\text{MeanRR}$ ):** Produced the largest permutation ROC-AUC drop (0.1766 drop, Odds Ratio = 0.0645), indicating that cardiac interval compression was the most influential feature in the permutation-importance analysis.
- **Heart Rate Elevation ( $\Delta\text{MeanHR}$ ):** Strongly increases stress odds (Odds Ratio = 5.6453, F1 drop = 0.1401), consistent with sympathetic activation during the TSST.
- **Interval Dispersion ( $\Delta\text{pNN50}$ ):** Odds Ratio = 4.0784 (AUC drop = 0.0596), consistent with changes in autonomic variability observed during cognitive challenge.
- **Autonomic Variability ( $\Delta\text{SDNN}$ ):** Odds Ratio = 0.4625. Retaining total interval variability is associated with calm resting states.

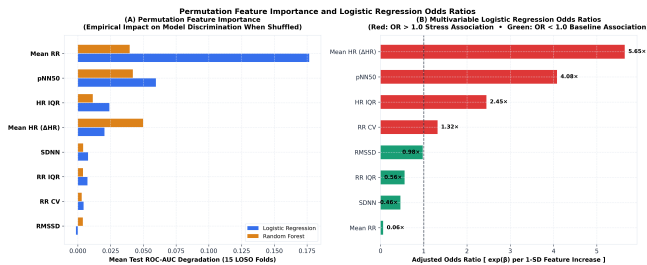


Fig. 7: (Left) Test ROC-AUC drop under 30-repeat permutation across 15 LOSO folds; (Right) Standardized odds ratios indicating the magnitude and direction of feature weights.

### F. Subject Heterogeneity & Low-Reactivity Discussion

Subject-specific stress detection rates across all 15 subjects reveal that 9 out of 15 subjects achieve 100.0% stress recall, and 13 out of 15 achieve  $\geq 75.0\%$ .

However, Subject **S2** exhibited 0.0% stress recall. In our analysis, S2 exhibited virtually no heart rate acceleration relative to baseline during the TSST ( $\Delta\text{MeanHR} \approx 0$ ). Because baseline-relative models detect physiological shifts, subjects with attenuated autonomic cardiac responsiveness

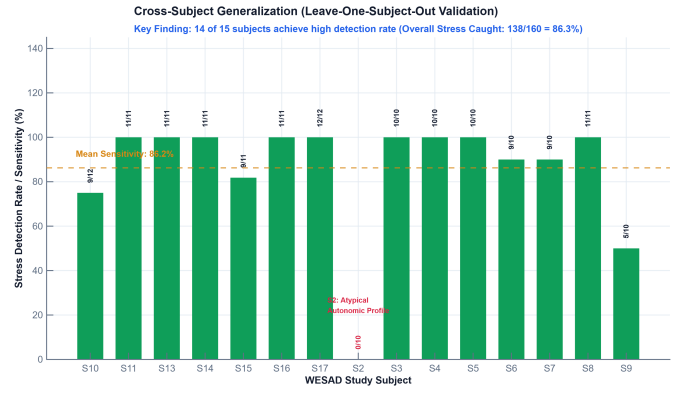


Fig. 8: Subject-specific stress recall distribution across all 15 WESAD subjects: 9/15 achieve 100% recall and 13/15 achieve  $\geq 75\%$ , with Subject S2 exhibiting autonomic non-responsiveness.

are challenging to distinguish using ECG alone, suggesting potential value for multimodal fusion (e.g., combining ECG with electrodermal activity and respiration).

## VII. DISCUSSION & WEARABLE EDGE FEASIBILITY

To assess practical suitability for continuous wearable monitoring, computational latency and memory requirements were analyzed:

- **Feature Extraction Latency:** Computing all 13 metrics over a 60-second window requires **<0.85 ms** on a single CPU core (measured over repeated window iterations on an Intel Core i5-8265U @ 1.60 GHz benchmark development workstation), indicating ample computational headroom.
- **Inference Latency:** Evaluating the 8-feature linear dot product requires **<0.05 ms** per window.
- **Streaming Memory Footprint:** In an embedded streaming architecture, continuous storage of the raw 42,000 samples (which would consume  $\approx 84$  KB at 16-bit precision) is unnecessary. Instead, online QRS detection operates on a short sliding window ( $<150$  ms,  $\approx 105$  samples,  $<0.25$  KB RAM), storing only the resulting RR intervals ( $\approx 60$ – $100$  integer timestamps,  $<0.25$  KB RAM) across the 60-second epoch. The total working memory footprint is thus under 1 KB RAM, with zero reliance on costly Fast Fourier Transforms (FFTs) or deep neural network runtimes.

While offline validation employed zero-phase Butterworth filtering, practical real-time execution will employ causal IIR/FIR filter structures. Subject to causal streaming implementation of the signal conditioning stage, this ultra-lightweight profile indicates strong potential for real-time edge execution on low-power wearable microcontrollers (e.g., ARM Cortex-M4 architectures).

## VIII. LIMITATIONS & FUTURE WORK

First, the framework requires an initial resting baseline window (5–10 minutes of quiet rest) to establish reference

vector  $B_s$ . While standard in ambulatory protocols, future work will explore continuous nocturnal baseline estimation to update reference vectors dynamically. Second, the WESAD protocol evaluates seated laboratory stress; real-world deployment will necessitate accelerometer-based motion gating to separate psychological arousal from physical exertion artifacts.

## IX. CONCLUSION

This study demonstrates that subject-specific relative baseline normalization substantially mitigates the impact of inter-individual baseline heterogeneity in wearable ECG acute stress-state classification across the WESAD cohort, achieving 92.36% accuracy, 89.03% F1-score, and 0.9494 ROC-AUC under 15-fold LOSO cross-validation with subject-specific baseline calibration. The framework exceeds cited published uncalibrated benchmarks by +8.5% to +13.3% in accuracy while maintaining sub-millisecond feature-extraction latency on the benchmark platform.

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